

Bayesian Optimisation — Week 1 Handover

Complete summary for the Week 2 LLM to continue from where Week 1 left off

Context

8 black-box functions are being maximised — one query per function per week — via a capstone project portal. Week 1 queries have been submitted. The next LLM should load the updated .npy files (n+1 rows each), re-fit the GP on expanded data, and generate Week 2 queries. All functions are maximisation problems. F3 and F6 have negative outputs by design — maximise toward zero.

Approach

Component	Choice	Notes
Surrogate	Gaussian Process (sklearn)	Re-fit from scratch each week on full dataset
Acquisition	UCB (F1-F4, F6-F8) / EI (F5)	EI for unimodal smooth surface; UCB elsewhere
Optimiser	Random scan + L-BFGS-B restarts	Bounds clipped 0.5% inward from [0,1] edges
Kernels	Per-function — see table below	ARD length-scales throughout (per-dim)
Kappa	High early (2.5-3.0), reduce over time	Reduce when best-y plateaus; raise if still flat
Override rule	Manual if GP contradicts strong correlation	Applied for F3 (x3 cliff), F8 (x5 direction)

Per-function detail

F1 2D Radiation detection may have two optima				GP TRUSTED
Kernel	Matern 5/2 + WhiteKernel	Acq fn	UCB k=3.0	
Best so far	y = 0.0000 (all 10 initial outputs near zero)			
Key drivers	No gradient signal. Pure exploration — probes unsampled top-right quadrant.			
Week 2 plan	If Week 2 also ~0: raise kappa, switch to structured grid to locate any peak.			
Submission	0.784742 - 0.801530			

F2 2D Noisy ML log-likelihood many local peaks				BLEND
Kernel	Matern 5/2 + WhiteKernel	Acq fn	UCB k=2.0	
Best so far	y = 0.6112 at (0.703, 0.927)			
Key drivers	Strong lead in top-right. GP x1 accepted; x2 overridden to exploit x2=0.93 lead. WhiteKernel essential — noisy surface.			
Week 2 plan	If y improves: continue pushing x2 toward 1. Check for second peak elsewhere.			
Submission	0.720632 - 0.970000			

F3 3D Drug discovery output transformed (minimise to maximise)				MANUAL OVERRIDE
Kernel	Matern 1/2 ARD (non-smooth cliff)	Acq fn	UCB k=2.0	
Best so far	y = -0.0348 at (0.493, 0.612, 0.340) [maximise toward 0]			
Key drivers	x3 dominant driver r=-0.57. Worst point at x3~1.0. GP failed to learn cliff from 15 pts — manual override kept x3 very low.			
Week 2 plan	If confirmed new best: push x3 toward 0.01 next round. All outputs negative.			
Submission	0.600000 - 0.720000 - 0.040000			

F4 4D Warehouse placement many local optima				GP TRUSTED
Kernel	Matern 5/2 ARD	Acq fn	UCB k=2.5	
Best so far	y = -4.026 at (0.578, 0.429, 0.426, 0.249)			
Key drivers	x1 r=-0.54, x4 r=-0.53 dominant. x3 nearly neutral r=-0.08. Best is outlier (2.7x rank 2). ARD weights dims unequally.			
Week 2 plan	GP explores different basin — do not over-exploit. Many local optima warrant coverage.			
Submission	0.407582 - 0.411176 - 0.347306 - 0.432593			

F5 4D Chemical process yield unimodal (single peak)				GP TRUSTED
Kernel	RBF ARD (smooth surface)	Acq fn	EI k=1.5	
Best so far	y = 1088.9 at (0.224, 0.847, 0.880, 0.879)			
Key drivers	Gradient: x1 down, x2/x3/x4 up. x4 strongest r=+0.57. EI predicts mu=1395 at query point.			
Week 2 plan	Continue EI exploitation every round. Push x1 lower; x2/x3/x4 toward 1. Single peak guaranteed.			
Submission	0.264706 - 0.995000 - 0.995000 - 0.995000			

F6 5D Cake recipe scoring outputs negative by design				BLEND
Kernel	Matern 3/2 ARD	Acq fn	UCB k=2.0	
Best so far	y = -0.714 at (0.728, 0.155, 0.733, 0.694, 0.056) [toward 0]			
Key drivers	x5 r=-0.58 (keep low), x4 r=+0.53 (keep high), x2 r=-0.33 (keep low). GP x2/x4/x5 accepted; x1 centred (neutral); x3 moderated.			
Week 2 plan	Confirm x5/x4 interaction. If new best: push x5 closer to 0, x4 closer to 1.			
Submission	0.500000 - 0.005000 - 0.900000 - 0.995000 - 0.005000			

F7 6D ML hyperparameter tuning			GP TRUSTED
Kernel	Matern 5/2 ARD	Acq fn	UCB k=2.0
Best so far	y = 1.365 at (0.058, 0.492, 0.247, 0.218, 0.420, 0.731) [2x next best]		
Key drivers	x5 r=-0.38, x1 r=-0.34, x6 r=+0.31. GP keeps x1~0, x6 high; explores different x2/x3. Best is big outlier.		
Week 2 plan	If GP query improves: tighten x1 toward 0 and reduce x5 further next round.		
Submission	0.005000 - 0.101223 - 0.524573 - 0.181478 - 0.365268 - 0.771330		

F8 8D Black-box (validation accuracy) local maxima acceptable			BLEND
Kernel	Matern 5/2 ARD (x1, x3 short priors)	Acq fn	UCB k=2.5
Best so far	y = 9.598 at (0.056, 0.066, 0.023, 0.039, 0.404, 0.801, 0.488, 0.893)		
Key drivers	x1 r=-0.63, x3 r=-0.62 co-dominant — keep both near zero. Good zone (x1<0.3 & x3<0.3) has only 3/40 initial pts.		
Week 2 plan	GP x1/x2/x3~0 accepted. x5->1 overridden (contradicts r=-0.17). x6/x8 high kept. Map good zone further.		
Submission	0.005000 - 0.005000 - 0.005000 - 0.400000 - 0.250000 - 0.850000 - 0.200000 - 0.880000		

Instructions for the Week 2 LLM

1. Load updated .npy files — each should have n+1 rows (one new result appended per function).
2. Verify shape before any analysis. Flag if unexpected row counts.
3. Re-fit GP with same kernel per function. Let hyperparameters re-optimize from new data.
4. Reduce kappa on functions where the Week 1 query improved the best observed value.
5. Keep kappa high (or raise further) on F1 if still flat. Switch to structured grid if second query also ~0.
6. F5: continue EI exploitation every round — push x1 lower, x2/x3/x4 toward 1.0.
7. F3: if x3=0.04 confirmed new best, push x3 toward 0.01 next round.
8. Run correlation analysis + GP acquisition for each function. Generate one submission string per function.
9. Submission format: values separated by ' - ' e.g. 0.123456 - 0.654321 (6 decimal places).
10. All 8 problems are maximisation. F3 and F6 have all-negative outputs — higher (less negative) is better.